“On the way with Python”

Lecture

06

Conditional Statements in Python

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Introduction

- Conditional statements (Decision-Making) are used to make decisions in a program. They allow the program to execute different blocks of code based on whether a condition is True or False.
- In Python, conditions are evaluated using comparison and logical operators, and the decision is made accordingly.

Why Conditional Statement are Important

Conditional statements are used to:

- Make decisions in programs
- Control the flow of execution
- Execute code selectively
- Handle different situations dynamically

Types of Conditional Statement

Python provides the following decision-making statements:

- if statement
- if-else statement
- if-elif-else statement
- Nested if statement
- Short-hand if (Ternary operator)

if Statement

The if statement executes a block of code only when the condition is true.

Syntax:

```python
if condition:
    statement
```

Example:

```python
age = 20
if age >= 18:
    print("Eligible for voting")
```

if Statement

Condition is True

```
number = 10
if number > 0:
    # code

# code after if
```

Condition is False

```
number = -5
if number > 0:
    # code

# code after if
```

if-else Statement

The if-else statement executes one block if the condition is true and another block if the condition is false.

Syntax:

```
if condition:
    statement
else:
    statement
```

Example:

```
num = int(input("Enter a number: "))
if num % 2 == 0:
    print("Even number")
else:
    print("Odd number")
```

If-else Statement

Condition is True

```
number = 10
if number > 0:
    # code

else:
    # code

# code after if
```

Condition is False

```
number = -5
if number > 0:
    # code

else:
    # code

# code after if
```

If-elif-else Statement

The if-elif-else statement is used to test multiple conditions sequentially.

python_if_elif_else.py

```python
if condition1:
    # code block 1

elif condition2:
    # code block 2

else:
    # code block 3
```

If-elif-else Statement

1st Condition is True

```
let number = 5
if number > 0 :
    # code

elif number < 0 :
    # code

else :
    # code


# code after if
```

2nd Condition is True

```
let number = -5
if number > 0 :
    # code

elif number < 0 :
    # code

else :
    # code


# code after if
```

All Conditions are False

```
let number = 0
if number > 0 :
    # code

elif number < 0 :
    # code

else :
    # code


# code after if
```

If-elif-else Statement

python_if_elif_else.py

```python
number = -5

if number > 0:
    print('Positive number')

elif number < 0:
    print('Negative number')

else:
    print('Zero')

print('This statement is always executed')
```

Nested if Statement

It is possible to include an **if** statement inside another **if** statement.

python_nested_if.py

```python
number = 5

# outer if statement
if number >= 0:
    # inner if statement
    if number == 0:
      print('Number is 0')

    # inner else statement
    else:
        print('Number is positive')

# outer else statement
else:
    print('Number is negative')
```

Short-hand if

The if statement can be simplified into a single line.

normal_if.py

```python
number = 10

if number > 0:
    print('Positive')
```

shorthand_if.py

```python
number = 10

if number > 0: print('Positive')
```

Short-hand if-else (Ternary Operator)

This technique is known as **Ternary Operators**, or **Conditional Expressions**. But Python doesn't have a ternary operator.

Syntax:

```
statement_if_true if condition else statement_if_fal
se
```

shorthand_if_else.py

```
number = 60

print('pass') if number >= 50 else print('fail')
```

normal_if_else.py

```
grade = 40

if grade >= 50:
    print('pass')
else:
    print('fail')
```

Python and Keyword

Returns **True** only if **both conditions are True**.

python.py

```python
age = 20
citizen = True

if age >= 18 and citizen:
    print("You are eligible to vote.")
else:
    print("You are NOT eligible to vote.")
```

Python or Keyword

Returns **True** if **at least one condition is True**.

python.py

```python
age = 17
citizen = True

if age >= 18 or citizen:
    print("You are eligible to vote.")
else:
    print("You are NOT eligible to vote.")
```

Python not Keyword

The **not** keyword is a logical operator, and is used to reverse the result of the conditional statement (True → False, False → True).

python.py

```python
is_sunny = False

if not is_sunny:
    print("It is cloudy or rainy today.")
else:
    print("It is sunny outside.")
```

“”“Thanks All””“